

Machine Learning: short introduction (trees and ANN), part 1

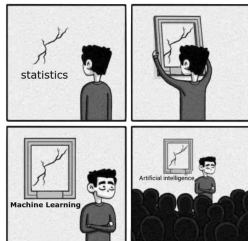
Sergey Korpachev on behalf of Dépôt

Outline

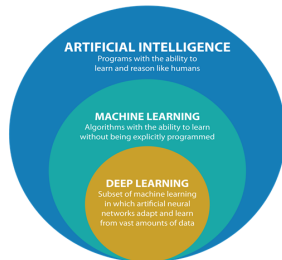
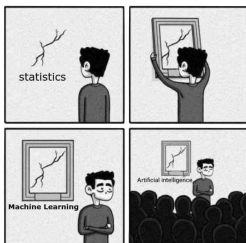
- Machine learning (ML)
 - Data
 - Features in ML
 - ML pipeline
 - Linear model
 - Decision tree
 - Neural network
 - Summary

Machine learning

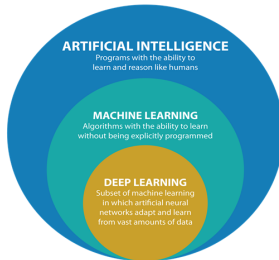
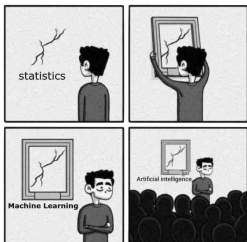
Machine learning —○ what is it?



Machine learning —○ what is it?

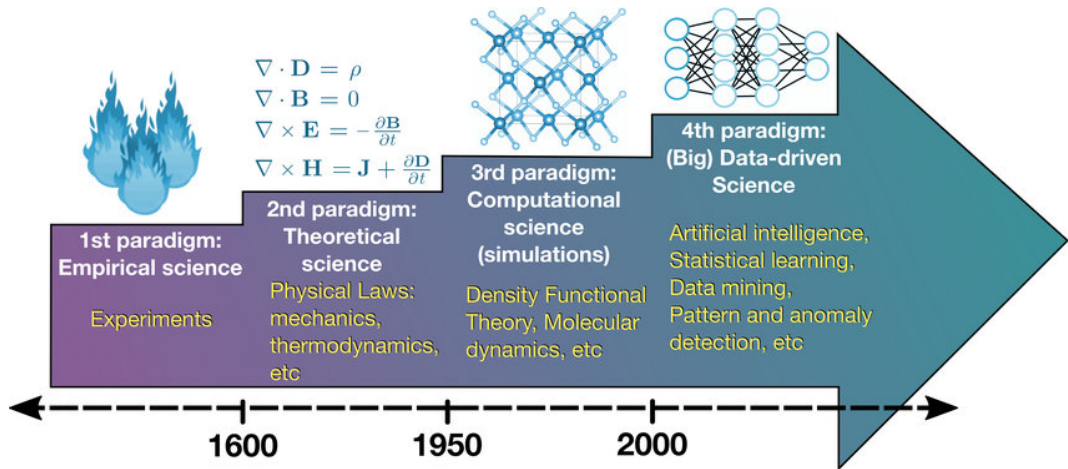


Machine learning —○ what is it?



Machine learning

—○ The Fourth Paradigm (Tony Hey, ...)



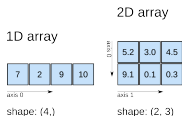
Data

Data

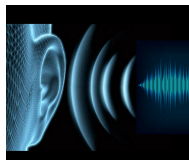
—○ what is data?

Anything can be data:

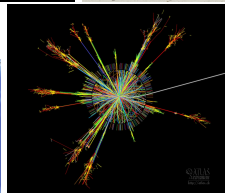
- 1 Numbers
- 2 Text
- 3 Images
- 4 Sound
- 5 Geomap
- 6 Particle collisions
- 7 Knowledge
- 8 You name it



3D array



18.
Shall I compare thee to a *Summer's day*?
Thou art more lovely and more temperate:
Rough winds do shake the darling buds of *May*,
And *Summer's* lease hath all too short a date:
Sometime too hot the eye of *Heaven* shines,
And often is his gold complexion dimm'd;
And every fair from *fair* *Season's* decline,
By chance, or *nature's* changing course, untrim'd;
But thy eternal *Summer* shall not fade,
Nor lose possession of that *fair* thou ow'st it,
Nor shall death brag thou *wast'st* it in his shade,
When in *eternal* lines to *time* thou grow'st;
So long as *men* can *breathe*, or *eyes* can *see*,
So long *lives* this, and *this* gives *life* to thee,



Features in ML

Features in ML —○ not HEP

Supervised learning

Classification

- cat, dog or muffin
- relevant or spam
- disease or not
- good or bad
- ...

Regression

- rent price
- temperature
- annual profit
- driving time
- ...

Don't forget about **unsupervised learning, reinforcement learning, semi-supervised learning** and so on.

Features in ML —○ HEP

Supervised learning

Classification

- b , c , uds jet
- π , K , μ particle
- $t\bar{t}$ or QCD event
- select or reject trigger candidate
- ...

Regression

- energy resolution
- pile-up mitigation
- ...

Don't forget about **unsupervised learning**, **reinforcement learning**, **semi-supervised learning** and so on.

Features in ML —○ ML challenges: slide 1 (A. Ustyuzhanin)

- Precise and fast particle tracking (single tracks, shower, jets)
- Particle identification
- Fast and accurate online data processing and filtering
- Anomaly detection (data quality monitoring, infrastructure monitoring)
- Detector design optimization (bayesian optimization, surrogate modelling)
- Data analysis (signal from background separation, ...)
- Simulation (speed-up simulation using generative models, simulator parameters optimization - tuning)
- ...

Features in ML —○ ML challenges: slide 2 (A. Ustyuzhanin)

Tracking system features

- Particle momentum
- Particle charge
- Track parameters
- Quality of track fit
- Number of track hits
- ...

RICH features

- Angle θ
- Quality of angle reconstruction
- Reconstructed particle type
- Reconstructed particle energy
- Light intensity
- ...

Features in ML —○ ML challenges: slide 3 (A. Ustyuzhanin)

Calorimeter features

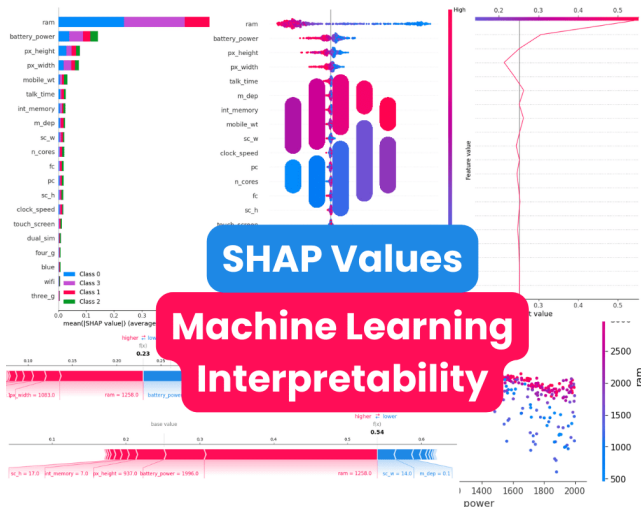
- Measured particle energy
- Shower parameters: length, width, ...
- Number of clusters in each layer Intensity of the clusters
- Intensity of the clusters
- Distance form track of the original particle
- ...

Features in ML —○ ML challenges: slide 4 (A. Ustyuzhanin)

Muon detector features

- Muon track parameters
- Quality of track fit
- Number of active layers
- Distance between the track and the active layers
- Length of shower
- ...

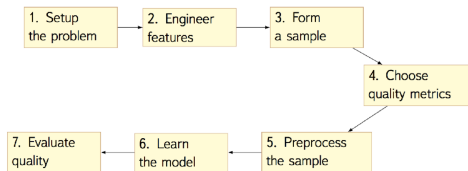
Features in ML — SHAP value



ML pipeline

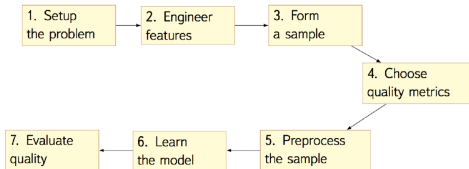
ML pipeline —○ what is ML pipeline?

Machine Learning Pipeline



ML pipeline —○ what is ML pipeline?

Machine Learning Pipeline



Machine Learning Pipeline

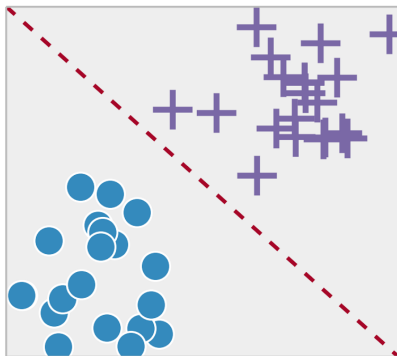


Linear model

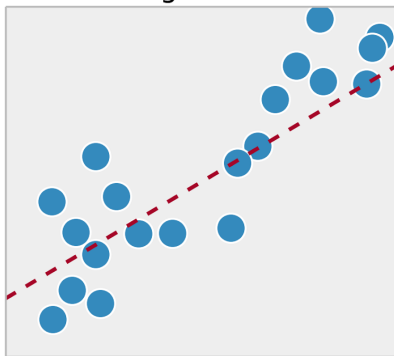
Linear model

—○ classification and regression

Classification



Regression



Linear model —○ regression

- $Y = \mathbb{R}$
- N objects with K real features: $D = \mathbb{R}^K$
- $a(\mathbf{x}, \boldsymbol{\theta}) = \theta_0 + \sum_{j=1}^K f_j(\mathbf{x}) \cdot \theta_j$
- Extend and reassign:
 $[1, f_1(\mathbf{x}), \dots, f_K(\mathbf{x})] \equiv \mathbf{x}$
 $[\theta_0, \dots, \theta_K] \equiv \boldsymbol{\theta}$
- Then $a(\mathbf{x}, \boldsymbol{\theta}) = \langle \mathbf{x}, \boldsymbol{\theta} \rangle$
- Minimization problem:
 - $\mathcal{L}(\boldsymbol{\theta}) = (\langle \mathbf{x}, \boldsymbol{\theta} \rangle - y)^2$
 - $Q(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N (\langle \mathbf{x}_i, \boldsymbol{\theta} \rangle - y_i)^2$
- Then we need to minimize $Q(\boldsymbol{\theta})$ by varying $\boldsymbol{\theta}$:
 - $\hat{\mathbf{a}} = \arg \min_{\boldsymbol{\theta} \in \Theta} Q(\boldsymbol{\theta})$

Linear model —○ classification

- $Y = \{-1, +1\}$
- N objects with K real features: $D = \mathbb{R}^K$
- $a(x, \theta) = \theta_0 + \sum_{j=1}^K f_j(x) \cdot \theta_j$
- Extend and reassign:
 $[1, f_1(x), \dots, f_K(x)] \equiv [1, x_1, \dots, x_K] \equiv x$
 $[\theta_0, \dots, \theta_K] \equiv \theta$
- Then $a(x, \theta) = \langle x, \theta \rangle$
- Minimization problem:
 - $\mathcal{L}(\theta) = [\text{sign}\langle x, \theta \rangle \neq y]$
 - $Q(\theta) = \frac{1}{N} \sum_{i=1}^N [\text{sign}\langle x_i, \theta \rangle \neq y_i]$
- Then we need to minimize $Q(\theta)$ by varying θ :
 - $\hat{a} = \arg \min_{\theta \in \Theta} Q(\theta)$